

SET 4: PRODUCT INVENTORY MANAGEMENT

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Product Inventory

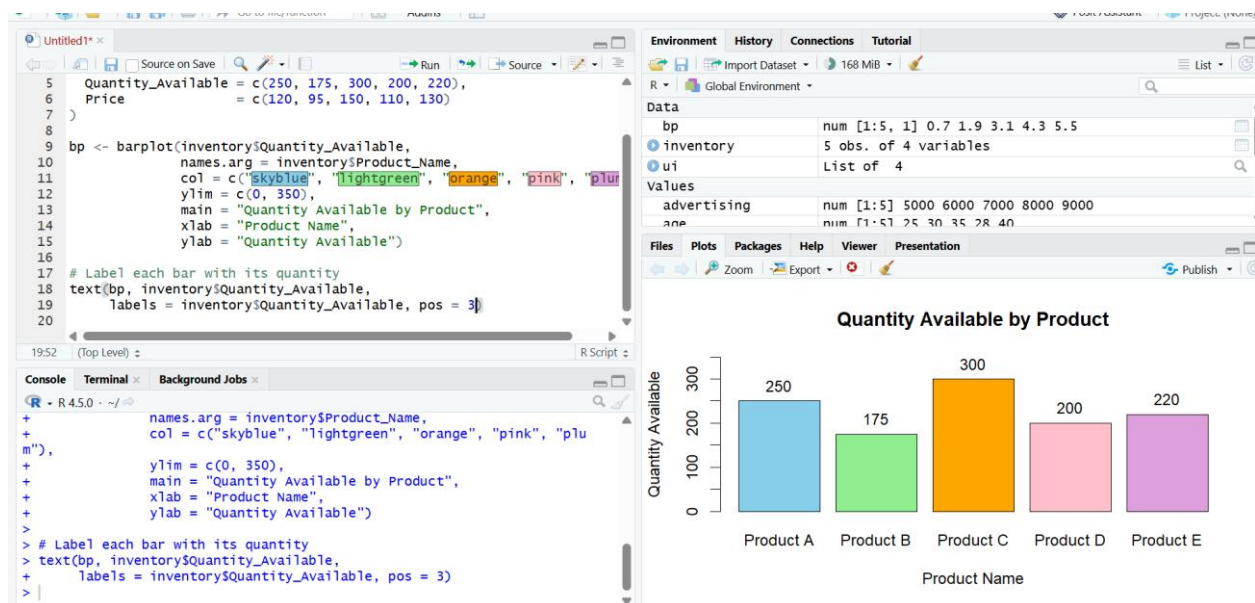
Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

PROGRAM 1: Bar Chart of Product Quantity

Question: Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

Code:

```
inventory <- data.frame(  
  Product_ID      = c(1, 2, 3, 4, 5),  
  Product_Name    = c("Product A", "Product B", "Product C",  
                      "Product D", "Product E"),  
  Quantity_Available = c(250, 175, 300, 200, 220),  
  Price           = c(120, 95, 150, 110, 130)  
)  
  
bp <- barplot(inventory$Quantity_Available,  
              names.arg = inventory$Product_Name,  
              col = c("skyblue", "lightgreen", "orange", "pink", "plum"),  
              ylim = c(0, 350),  
              main = "Quantity Available by Product",  
              xlab = "Product Name",  
              ylab = "Quantity Available")  
  
# Label each bar with its quantity  
text(bp, inventory$Quantity_Available,  
      labels = inventory$Quantity_Available, pos = 3)
```



Output 1: Quantity Available by Product

Observation: The bar chart shows the quantity available for each product. Product C has the highest stock (300 units) and Product B has the lowest (175 units). Product A (250), Product E (220) and Product D (200) lie in between. The total inventory is 1145 units, so Product B may need restocking first.

PROGRAM 2: Stacked Bar Chart of Quantity by Category

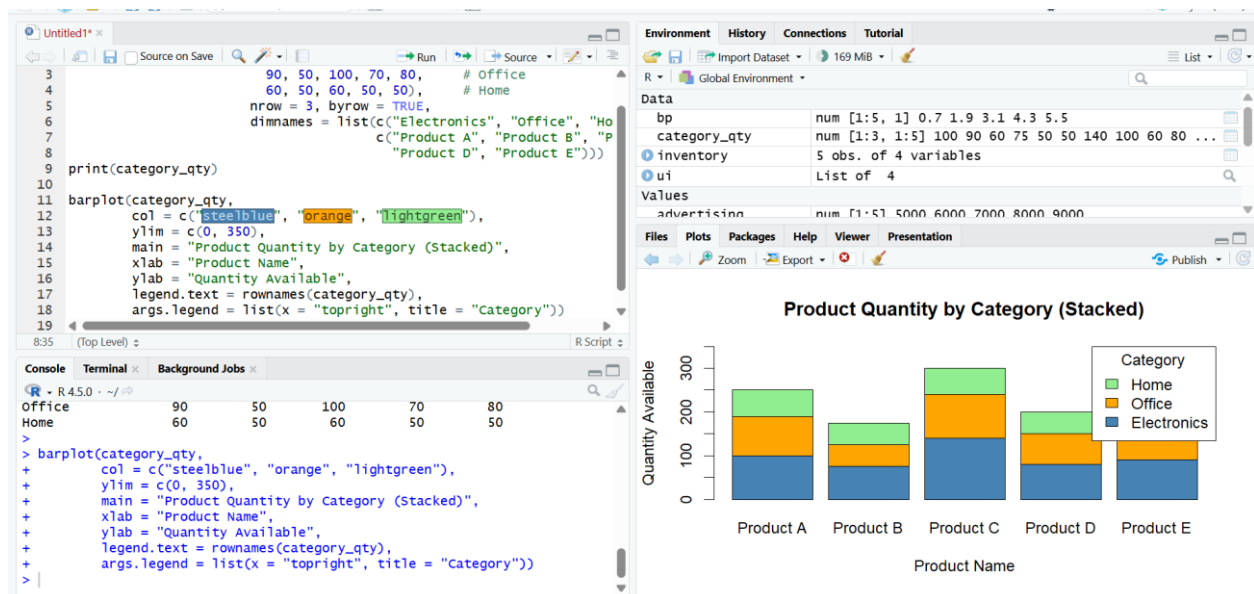
Question: Generate a stacked bar chart to show the quantity of each product within different product categories.

Code:

```
# Rows = categories, columns = products
category_qty <- matrix(c(100, 75, 140, 80, 90,      # Electronics
                        90, 50, 100, 70, 80,      # Office
                        60, 50, 60, 50, 50),      # Home
                      nrow = 3, byrow = TRUE,
                      dimnames = list(c("Electronics", "Office", "Home"),
                                     c("Product A", "Product B", "Product C",
                                       "Product D", "Product E")))

print(category_qty)

barplot(category_qty,
        col = c("steelblue", "orange", "lightgreen"),
        ylim = c(0, 350),
        main = "Product Quantity by Category (Stacked)",
        xlab = "Product Name",
        ylab = "Quantity Available",
        legend.text = rownames(category_qty),
        args.legend = list(x = "topright", title = "Category"))
```



Output 2: Stacked Bar Chart of Quantity by Category

	Product A	Product B	Product C	Product D	Product E
Electronics	100	75	140	80	90
Office	90	50	100	70	80
Home	60	50	60	50	50

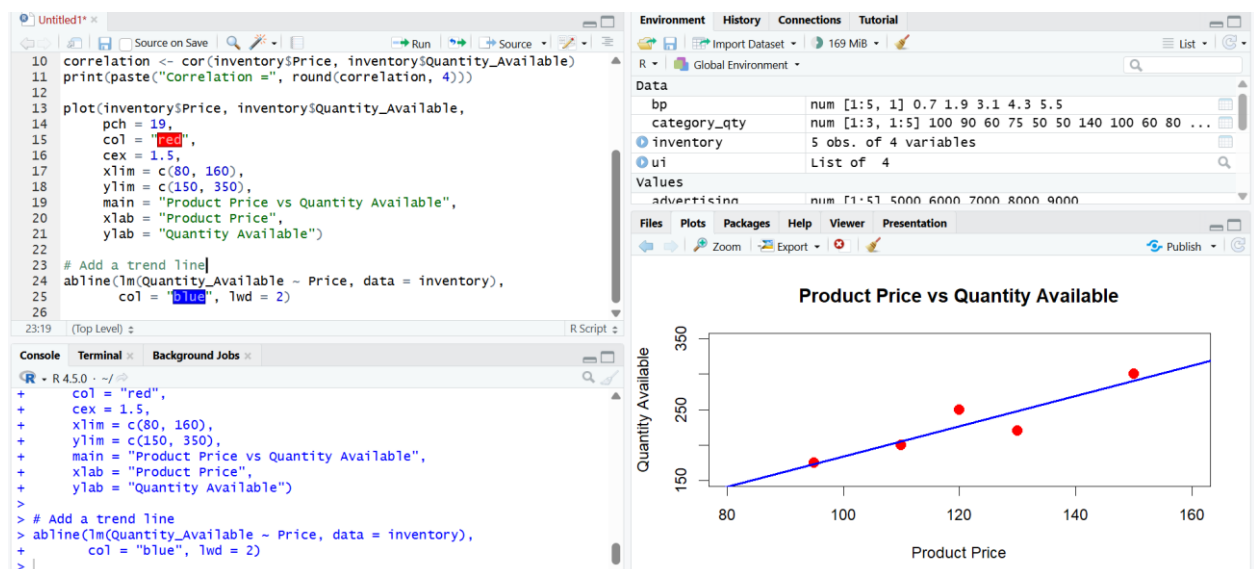
Observation: Each bar shows the total stock of a product, split into Electronics, Office and Home. Electronics contributes the most in every product, with 485 units in total, followed by Office with 390 units and Home with 270 units. Product C has the tallest bar (300), and Product B has the shortest (175). This helps to see both the total quantity and the category mix of each product in one chart.

PROGRAM 3: Scatter Plot of Price vs Quantity Available

Question: Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

Code:

```
inventory <- data.frame(  
  Product_ID      = c(1, 2, 3, 4, 5),  
  Product_Name    = c("Product A", "Product B", "Product C",  
                      "Product D", "Product E"),  
  Quantity_Available = c(250, 175, 300, 200, 220),  
  Price           = c(120, 95, 150, 110, 130)  
)  
  
# Calculate the correlation coefficient  
correlation <- cor(inventory$Price, inventory$Quantity_Available)  
print(paste("Correlation =", round(correlation, 4)))  
  
plot(inventory$Price, inventory$Quantity_Available,  
     pch = 19,  
     col = "red",  
     cex = 1.5,  
     xlim = c(80, 160),  
     ylim = c(150, 350),  
     main = "Product Price vs Quantity Available",  
     xlab = "Product Price",  
     ylab = "Quantity Available")  
  
# Add a trend line  
abline(lm(Quantity_Available ~ Price, data = inventory),  
       col = "blue", lwd = 2)
```



Output 3: Product Price vs Quantity Available Scatter Plot

```
[1] "Correlation = 0.9191"
```

Observation: The correlation between product price and quantity available is 0.9191 (about 0.92). The positive sign means that, in this dataset, products with a higher price also tend to have more stock available. Since the value is close to 1, the relationship is strong, and the points lie

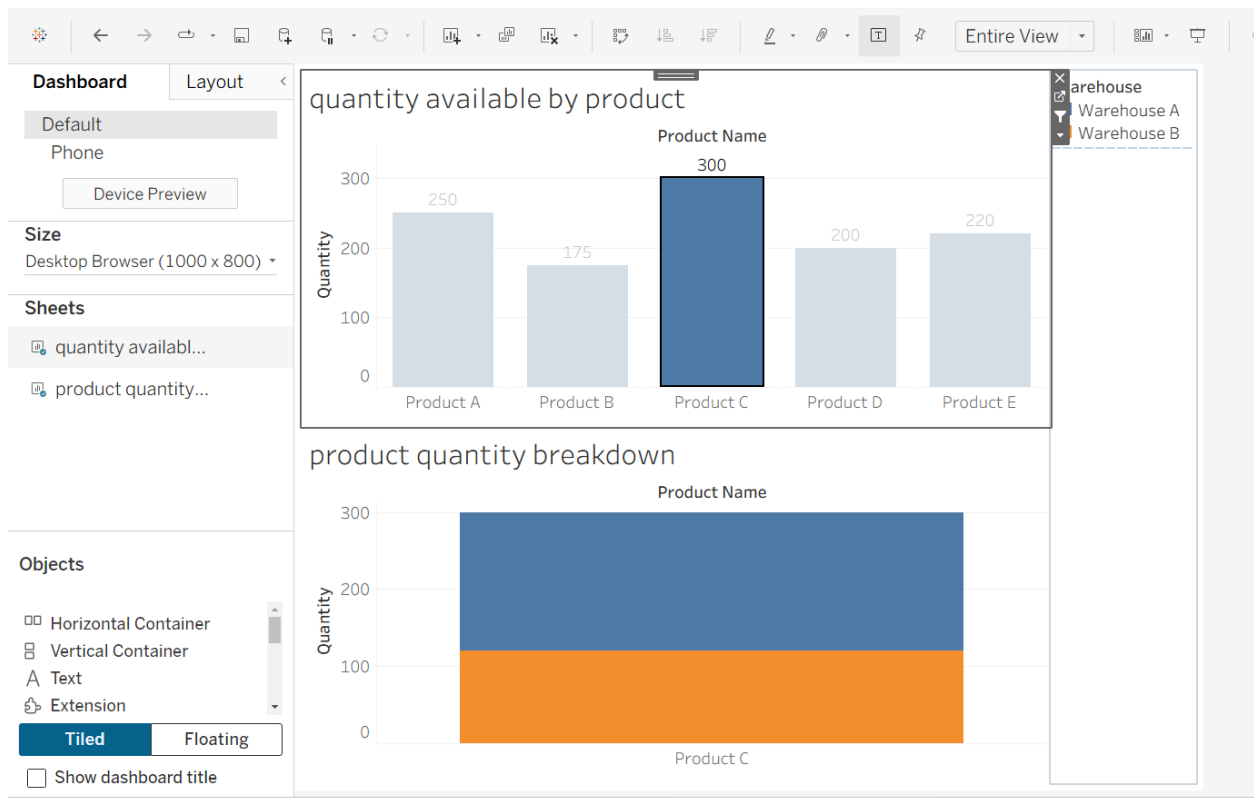
close to the upward trend line (slope about 2.14, i.e. roughly 2 more units for every 1 unit increase in price). This result only describes these 5 records; it does not prove that price causes higher stock, and the price values are assumed sample data, so a larger real dataset would be needed for a reliable conclusion.

PROGRAM 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Steps to create the dashboard in Tableau:

1. Save the Product Inventory data in long format (Product ID, Product Name, Category, Quantity Available, Price) as an Excel or CSV file. Each product has three rows, one for each category (15 rows in total).
2. Open Tableau, click Connect to a File (Microsoft Excel or Text file), select the file and open the Sheet 1 tab.
3. Right-click Product ID in the Data pane and choose Convert to Dimension, so it is not summed as a measure.
4. Bar chart: rename the sheet as "Product Quantity". Drag Product Name to Columns and Quantity Available to Rows (SUM). Set the Marks type to Bar, click Show Mark Labels to display the quantities, and edit the title and axis labels.
5. Stacked bar chart: open a new sheet and rename it as "Quantity by Category". Drag Product Name to Columns and Quantity Available to Rows. Drag Category to Color so that each bar is stacked by category and the legend appears. Set the Marks type to Bar, show the mark labels, and edit the title and axis labels.
6. Click New Dashboard, rename it as "Product Inventory Dashboard", and set the size to Automatic.
7. Drag the "Product Quantity" sheet and the "Quantity by Category" sheet from the Sheets list into the dashboard area. Show the dashboard title from the Dashboard menu.
8. Interactive filter: on the Quantity by Category sheet, drag Category to Filters, select all categories, and choose Show Filter. In the filter's drop-down, choose Apply to Worksheets > Selected Worksheets, tick both sheets and click OK. Also click the Use as Filter icon on the stacked bar chart so that clicking a product bar filters the other chart.
9. Test the dashboard by selecting different categories and products and check that both charts update. Take a screenshot of the final dashboard.



Output 4: Tableau Product Inventory Dashboard

Observation: The dashboard shows the quantity of each product (bar chart) and the category-wise split of the quantity (stacked bar chart) side by side. Using the Category filter, or by clicking a bar, the user can view the data of a single product or category such as Electronics, Office or Home and see how the stock changes. This makes it easy to compare products and categories interactively.

PROCEDURE

1. Create vectors and a data frame (inventory) in R containing Product ID, Product Name, Quantity Available and Price.
2. Create the bar chart of Quantity Available for each product using the `barplot()` function. Add the title, axis labels and bar labels using `text()`.
3. Create a matrix of quantities by category and draw the stacked bar chart using `barplot()` with a legend.
4. Create the scatter plot of Price and Quantity Available using `plot()`, add the trend line with `abline(lm())` and calculate the correlation using the `cor()` function.
5. Import the Product Inventory dataset into Tableau and convert Product ID to a dimension.
6. Create the bar chart (Product Name vs Quantity) and the stacked bar chart (Product Name vs Quantity, coloured by Category) in separate worksheets.
7. Combine both worksheets in a Tableau dashboard, add the Category filter and test the interactivity.

RESULT

Thus, the product inventory data was successfully visualized using a bar chart, a stacked bar chart, a scatter plot and a Tableau interactive dashboard. The scatter plot was used to examine the relationship between product price and quantity available, and it showed a strong positive correlation ($r = 0.9191$) in the given dataset.